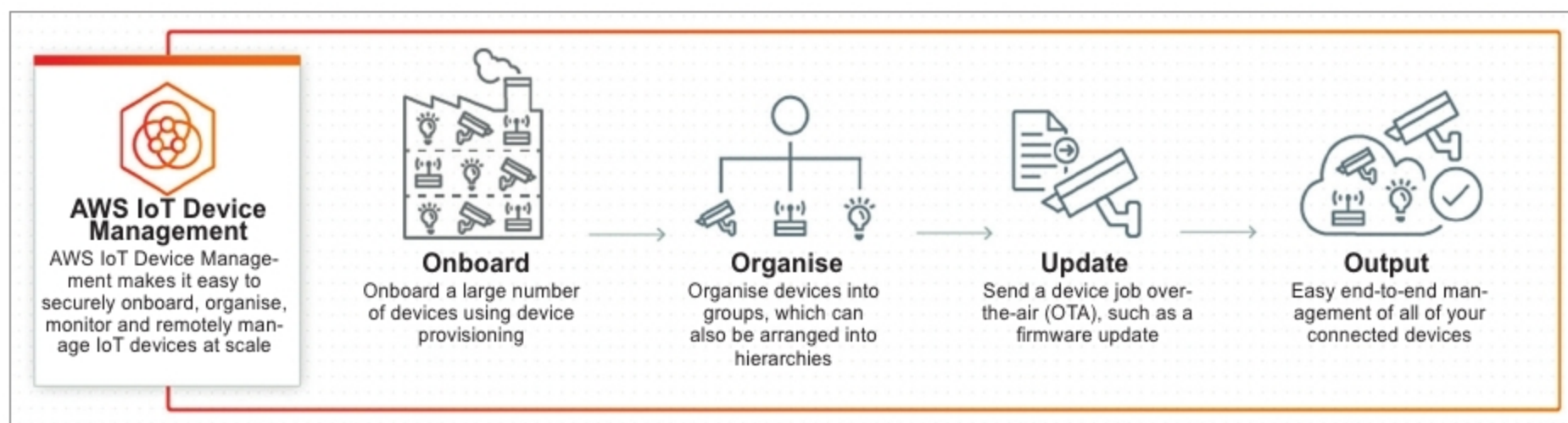




Working of AWS IoT device management (Credit: <https://aws.amazon.com>)

reduce bandwidth cost. These also provide great performance, ownership over data and lower maintenance cost.

Edge devices run a Web-based dashboard that end-users can access to monitor the flow of data, so they can decide how various systems in demonstration and devices are running, and get notified by alarms. A large amount of data can be represented in the form of a graph for any desired range of time. Each point on the graph represents a record that can be found by searching the database, which stores a large quantity of data.

AWS IoT device management: This is critical across industrial, consumer and commercial applications such as an industrial and connected home. It enables customers to manage large and diverse device fleets such as operational technology systems, cameras, machines, vehicles, appliances and more.

It enables monitoring of usage and performance metrics for devices across industrial sectors such as manufacturing, oil and gas, and mining. It allows monitoring metadata and policy changes with service alerts to inform about any adjustments to the devices' configuration. It also allows detection of any unusual behaviour across a device and to take mitigating actions.

It provides secure onboarding, organising, monitoring, troubleshooting and sending of firmware updates over-the-air (OTA). It enables adding device attributes like device name, type and manufacturing year, certificates and access policies to the IoT Registry. Then, it assigns them to devices, and makes

connected devices ready for service quickly. This helps to quickly search and find any IoT device across the entire device fleet in near real-time.

It is easy to find devices based on a combination of attributes like device ID, device state and type, so that action can be taken, in addition to troubleshooting. Actions such as reboots, factory resets and security patches can also be remotely executed.

Data management challenges

With time, the number of IoT devices will increase, thus increasing the challenges for real-time processing and analysis to reduce time for storage. Space has to be optimised for metadata like user IDs and passwords to ensure enough space for new information.

Functions such as adaptive maintenance, predictive repair, security monitoring and process optimisation rely on real-time data. Selecting the right tools is a challenge because integration between different sensors should be proven and compatibilities confirmed. When there is no connection, devices must still gain insights, make decisions and prepare for data distribution.

There are important factors behind an IoT device data management platform, including interoperability, scalability, security and standards offered by software technologies to build IoT products. It is important to protect data from unauthorised access and tampering. Organisations need to be compliant with national rules and regulations on securing data.

IoT device data also needs to be checked for quality. Having many different devices connected directly to cloud services presents a huge attack surface, which can be mitigated by channelling data through a secure gateway device.

Potential of IoT device management

IoT device management software enable users to track, monitor and manage physical IoT devices to ensure they work properly and securely after deployment. These tools often allow them to push software and firmware updates to devices and troubleshoot problems remotely.

They also provide permissions and security capabilities to ensure each device is protected from vulnerabilities. These solutions are mainly used by IT administrators to track performance, security and the overall state of each connected device.

Exploring IoT data from the field will give a better idea of how the product functions in a user's day-to-day life. IoT data management lets the user optimise smart algorithms by looking at sensory data and the points in time when a user made a change. Then, the user can redesign or retrain the product to offer a better user experience. **END** 🐧

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